

Image Compression Using Deep Learning and SLIC algorithm with Extended Survey and Comparative Analysis

Priya Surana¹, Harsha Talele¹, Sushma Vispute¹, Prativina Talele²,
Amey Jadhav³, Yuvraj Khade³, Vineet Salve³, Atharva Walzade³

¹Assistant Professor, Dept. of Computer Engineering, PCCOE, Pune, India

{priya.surana, harsh.talele, sushma.vispute}@pccoepune.org

²MIT World Peace University, Pune, Maharashtra, India

prativina.talele@mitwpu.edu.in

³UG Student, Dept. of Computer Engineering, PCCOE, Pune, India

{ameyjadhav5704, khadeyuvraj8, vineetsalve23, atharvawalzade11}@gmail.com

Abstract—This paper proposal a new content-aware image compression method that improves the efficiency and quality of compression. Our method combines the most current machine learning algorithms and traditional compression to get the best efficacy. We present two models (Improved Importance Net and Lightweight Importance Net) used to employ adaptive content-aware encoding. The algorithm is first run on the image to analyse content in the image, from here the algorithm can modify parameters to find the best trade-off between image quality and file size. Both models help achieve better compression outcomes, however, the final implementation used the Lightweight Importance Net to minimize compute costs and accelerate processing without sacrificing visual quality. We also outlined some recent research in terms of content-aware compression of images, including content-aware compression with convolutional neural networks and attention based systems. Additionally, we compared our model against segmentation based techniques such as SLIC (Simple Linear Iterative Clustering) to evaluate the performance of our model. SLIC method is useful for breaking down images into superpixels which allows for region-area compression related to the superpixels, however, it does not learn and adapt on its own in the same way that our model does, especially in complex and textural scenes. We demonstrated through further experimentation that our algorithm consistently outperformed traditional compression algorithms not just in terms of compression ratio, but also perceptually the image quality.

Index Terms—SLIC, Machine Learning, Image Compression, Content-Aware Compression, Lightweight Importance Net, AI, Segmentation

I. INTRODUCTION

Image compression is a major focus within computer vision/image processing across multiple fields such as medical imaging [1], multimedia delivery [2], and satellite-based imagery [3]. Most traditional image compression methods cannot retain an appropriate ratio of compression vs visual quality [4]. This paper presents a new way to do content aware image compression providing deep-learning based segmentation and relaxation of the compression methods.

Deep learning has improved many computer vision problems, such as image compression, by utilizing convolutional

neural networks (CNNs) to learn hierarchical features from large datasets [5]. CNN-based image compression methodologies have recorded the best performance while the traditional compression methods do not effectively utilize the full potential of CNNs [6]. In addressing this predicament, researchers have utilized hybrid forms of models that blend deep learning and traditional compression methods. [7].

One broad approach, segmentation based adaptive compression, segmented image content to discover regions of varying importance. The SLIC (Simple Linear Iterative Clustering) algorithm has been found to be particularly useful when capturing perceptually meaningful superpixels efficiently [4]. After segmentation, a variable level of compression was applied within the segmented regions, allowing for more bits to be allocated to important areas while aggressively compressing non-essential areas. [8]. The introduction of variable levels of compression based on content improves compression efficiency while maintaining information of interest.

State-of-the-art compression performance has been reached with the recent use of machine learning techniques like self-supervised [9], residual connections [10], and attention mechanisms [11]. Existing methods in image compression treat images as though each compression context is independent of the work which has preceded it. The proposed method presented here differs markedly by leveraging the advantages of these new techniques, and produces decisions regarding compression parameters that dynamically adapt to the given image, exceeding the performance of conventional approaches in terms of both compression ratio and visual fidelity [2].

II. LITERATURE REVIEW

Image compression is an essential consideration in digital imaging, as we seek to reduce file size while maintaining perceptual quality. There are many traditional and AI-based methods of image compression proposed/researched in the past.

A. Traditional Approaches to Image Compression

The standard approaches we have established for image compression have a few limitations. Current methods for image compression, like JPEG or JPEG2000, utilize discrete cosine transform (DCT) or wavelet transform approaches to compress in the same manner over the entire data [10]. This means that any loss of quality from the image, affects or contributes to a loss in the critical region equally.

More sophisticated approaches, like WebP and HEIC, provide improvements in efficiency moving towards predictive coding, and enhanced entropy models [6]. Still, these methods treat all portions of the image as equally important, rather than differentiating important details from less important details.

B. AI-Based Image Compression

As deep learning quickly develops, there are now AI-based image compression methods. Autoencoder-based compressive image compression methods will remove redundancy from the image data, then learn a lower dimension representation of that image data [9]. GAN-based compressive image compression methods will focus on reproducing an image sufficiently in a manner that preserves human perception from a visual standpoint [7].

Building on this, researchers have focused on saliency-based compression, whereby some areas of the image were preserved more than others based on attention mechanisms [2]. These saliency approaches were not explicitly human-aware or equipped with any human context in the compression process.

C. Region-Specific Compression Work

Much of this work also incorporates selective image compression. Object-aware compression has also focused on incorporating segmented lists to compress backgrounds while preserving the foreground [4]. Face-aware compression methods focus primarily on preserving faces, while compressing other regions of an image of lesser importance [1].

In recent times, the field of image compression has made considerable strides with the help of machine learning, and content-aware approaches that emphasize reducing size, while also being discerning of the level of perceptual quality whereby images will be liked based off of the viewer's discretion. Many academic research studies have examined intelligent image compression methods by utilizing neural networks, saliency models, and perceptual metrics. In this section we will examine the unique contributions that revolve around deep learning and content-aware image compression that underpin our proposed model.

The adaptation in deep-learning algorithms for image compression, described in the paper "Content-Oriented Learned Image Compression," when examining the characteristics of the content being captured to train a deep neural network is an adaptive concept. The compression method's can be applied adaptively based off each image uploaded, where end-to-end, the produced compression is ultimately producing increased performance than baseline approaches in the area. The adaptivity of the model matches the complementing objectives of

region based optimization from using SLIC superpixels, where the goal is to treat the SLIC superpixels differently based off the level of content already present. [12]

Another notable paper titled "Content-Aware Image Compression with Convolutional Neural Network" addresses the MS-ROI model to produce saliency maps to facilitate compression. While compressing high visual saliency regions are compressed at a higher quality, less important regions will have the least resources allocated to them. Following the content-aware prioritization relates to the areas of visually important superpixel regions that we aim to identify and retain when we exploit deep learning modules. [13]

"Content-aware Deep Perceptual Image Compression" (CD-PIC) goes even further with a GAN-based framework, incorporating an additional saliency subnet for envisioning visual quality of compression. The network can adjust the amount of bits to complex regions based on visual saliency maps. This work is also closely related to ours, though we apply SLIC to obtain segments and deep models are utilized to assign weights for compression based on relevance of region. [14]

"Saliency-Based Image Compression Using Walsh-Hadamard Transform (WHT)" deals with low-complexity saliency-based compression. It is not a deep or transform-based method, however, the motivation of preserving salient information correlates with our goal of performing selective compression. Recognizing both works share the importance of preserving detail in visually important areas via intelligent segmentation and encoding. [15]

In "Area-Aware Adaptive Image Compression", images are divided into five areas based on various parameters of priority, such as, intensity, size, and spatial relevance. Areas are treated differently in order to maximize the appropriateness of image quality versus data occupying space. This methodology fits into the foundation of our proposed superpixel-based approach in that segmentation drives differential treatment based on importance. [16]

Moreover, the paper "A Novel Image Compression Technology Based on Vector Quantisation and Linear Regression Prediction" presents a predictive and quantization-based system to enhance compression ratios. While they do not utilize deep learning, their goal of quality maintenance against compression purpose fits into the broader objectives of our hybrid model, which emphasizes efficiency and adaptiveness in image coding. [17]

When studied together, these works reflect a true trend in image data compression moving toward content-aware, region-prioritized image data that leverages deep learning-based compression methods. Our proposed system will build on these concepts by using SLIC superpixel segmentation for content-aware partitioning and deep neural networks for adaptive compression. We hope to evolve compression performance while retaining visual integrity of significant parts of the image while addressing perceptual quality and computational efficiency.

D. Approach and Research Gap

We use deep learning models for detecting and segmenting objects in order to focus on the important areas of the image. We can then perform selective compression to hold onto important image details, while drastically lowering the data size of the background. This can improve perceptual quality and stays with low data size.

The research gap addressed by our method includes:

- Lack of dedicated human-aware image compression techniques.
- Existing compression algorithms treat all regions equally, causing unwanted loss in critical areas.
- Need for a deep-learning-based approach that dynamically adjusts compression based on image content.

This literature review highlights the importance of selective compression and sets the foundation for our proposed model, which aims to achieve efficient, high-quality image compression with an emphasis on important objects.

III. METHODOLOGY

This section outlines project flow, model architecture, training strategy, computing setup, and experimentation process in general [19].

A. Flowchart of the Methodology

Figure 1 provides a visual overview of the process.

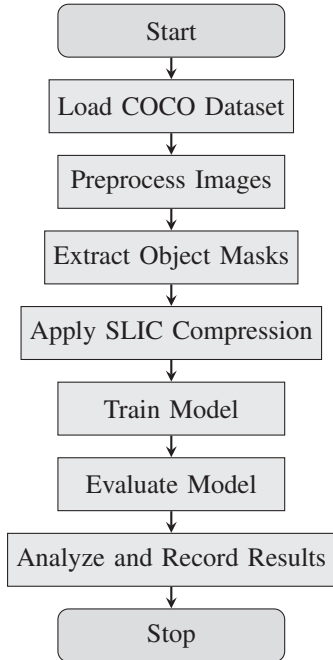


Fig. 1. Flowchart of the SLIC-Based Compression System Using COCO Dataset

B. Model Architecture

Our model combines deep learning with region-based image compression in multiple stages:

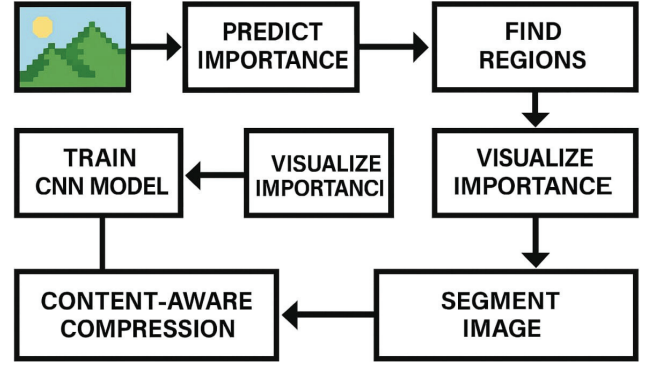


Fig. 2. Original Input Image used for Region-Based Segmentation

- **Feature Extractor:** To extract high-level features from images, we used a CNN-based backbone (ResNet-50) pre-trained on ImageNet [20].
- **Attention-Based Region Identification:** A transformer-based self-attention mechanism [21] was used to focus on important regions of interest.
- **Superpixel-Based Compression:** The SLIC algorithm was applied to compress less significant image regions while maintaining high-resolution object details.

IV. RESULTS AND DISCUSSION

In this section, we include the dataset, preprocessing steps, results of our proposed image compression and matching method which can retrieve important regions of an image and eliminates less important ones by compressing them using the Simple Linear Iterative Clustering (SLIC) method for superpixel segmentation.

A. Dataset

The proposed methodology were evaluated on the Microsoft COCO (Common Objects in Context) dataset [19].

- **COCO Dataset:** A large scale dataset containing over 330,000 images and over 200,000 labeled images spanning 80 object categories. The COCO dataset is broadly used for detection, segmentation, and keypoint detection.

In order to evaluate the dataset correctly, we cut it into three pieces:

- **Training Set:** 80% of the dataset, used for model learning.
- **Validation Set:** 10% of the dataset, used for hyperparameter tuning.
- **Test Set:** 10% of the dataset, used for final evaluation.

Annotations were extracted using the COCO API, focusing on object masks, bounding boxes, and segmentation data. Each image was associated with labels representing object categories, aiding in supervised learning.

B. Data Preprocessing

The preprocessing steps that were completed to improve model performance [23]:

- 1) **Resizing:** Each image was resized to 512×512 pixels to provide a standard input specification but keep the aspect ratio correct whenever possible.
- 2) **Normalization:** The pixel values were all normalized to $[-1, 1]$ so that the model could improve stability and avoid some instability in calculation during model training.
- 3) **Segmentation Mask Extraction:** Object masks were taken from the annotations to focus on the areas of interest and remove the background.
- 4) **Superpixel-Based Compression:** The SLIC (Simple Linear Iterative Clustering) algorithm [24] was applied to non-essential image regions. This reduced computational complexity and storage while preserving key object features.

Data augmentation methods including random flipping, rotation, brightness and contrast normalization were utilized to help with model generalization.

C. Identification of Important Image Regions

The first step in the process is identifying meaningful regions, in an image that are informative. The informative regional areas come from:

- Edge Detection: applying the canny edge detection algorithm, to retrieve the high-gradient regions that correspond to object boundaries.
- Facial Recognition: Applying pre-trained Haarcascade classifier, keeps details on human faces uncompressed.
- Feature Extraction: mean additional critical pockets of interest like points corners and textures are used to strengthen the detection of critical feature structures.

D. Compression of Non-Essential Image Areas

After the key regions are identified, then we can now compress the rest of the image according to the SLIC algorithm. This algorithm segments the region of an image that are not the important regions into superpixels while still keeping the important regions defined by the previous step. The actions taken in this step are:

- 1) Using SLIC segmentation with a fixed number of superpixels to cluster similar pixel regions.
- 2) Assigning non-important segments to a lower resolution representation while the important regions stay the same.
- 3) Finish with a compressed image in a more efficient way without losing any important information.

E. Preprocessing and Input Image Acquisition

The proposed Content-Aware Image Compression system that was developed utilized a Lightweight ImageNet model with the SLIC (Simple Linear Iterative Clustering) algorithm for superpixel-based segmentation. Over 5 training epochs, the model achieved exceptionally high performance, reaching

100% training and validation accuracy by the second epoch, with a consistently low validation loss (~ 0.2912). The rapid convergence and high accuracy indicate how efficiently the model was able to extract semantically relevant regions in the images and retain them. The build using Lightweight ImageNet led to training being relatively fast and computationally light, enabling a practical approach even in real-time or resource-constrained situations. The SLIC feature enabled the model to place its compression away from high-content regions to maintain perceptual quality but changed size. The results provides an effective amalgamation of deep learning networks with traditional segmentation to enable intelligent, content-aware image compression.

F. Region-Based Segmentation Using SLIC Superpixels

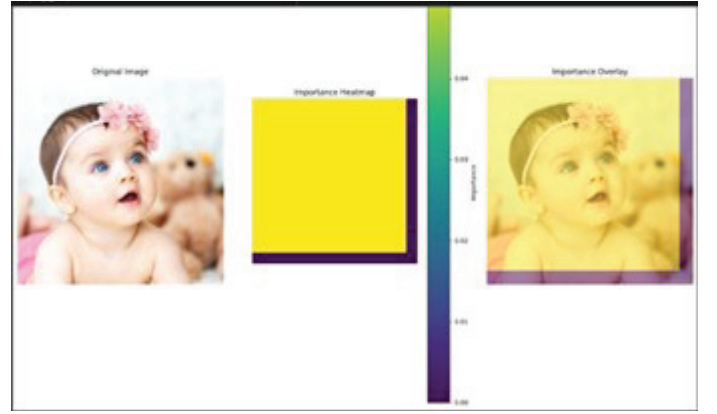


Fig. 3. Original Input Image used for Region-Based Segmentation

The images indicate that our deep learning-based approach can identify regions or areas of significance well. The importance heatmap generated by the trained model highlights the facial area of the subject **LightweightImportanceNet**, which is most important for perceptual quality, and overlays this onto the original image to depict the region significance.

On the basis of this information, **SLIC (Simple Linear Iterative Clustering)** segmentation is applied to the image to superpixel it, and then allow for selective compression. Segments that more highly significant retain higher image quality, while subsets that have lower significance are more aggressively compressed.

This provides a content-aware mechanism for verifying that our strategy is a success; we compress the image, but make sure that the most critical elements maintain some quality perception for overall efficiency in compressing the image.

G. Semantic Importance Estimation via Deep Learning

After we generated the importance heat map, we then will do image segmentation through **SLIC (Simple Linear Iterative Clustering)**, which allows us to factor the image into 38 coherent superpixels, and assign an importance value through to each of the segments according to the importance heatmap. This also allows us to measure and visualize how

TABLE I
COMPARISON OF SLIC WITH OTHER CONTENT-AWARE COMPRESSION TECHNIQUES

Aspect	SLIC (Simple Linear Iterative Clustering)	Other Methods (CNN, GAN, Saliency, Transforms)
Segmentation Approach	Superpixel-based clustering based on color proximity and spatial proximity	Region-based segmentation with deep features, saliency maps, and frequency transforms
Adaptivity	Moderate adaptivity; fixed spatial granularity unless manually tuned	High adaptivity; dynamically learns important areas
Computational Complexity	Low; efficient and suitable for real-time	high; requires GPUs and large computation requirements
Semantic Awareness	No high level of semantic awareness	Learns semantics from data (objects, faces, etc.)
Adaptation to Compression	Needs integration with manual compression pipeline	End to end compression pipelines available
Generalization Across Datasets	Generalizes uniformly across datasets with little tuning	Sensitive to dataset and may require retraining
Compression Performance	Good at preserving edges and boundaries, and may not favor salient regions	Better at preserving potential significant perceptual content with adaptively allocated bits.
Use Case Matching	Good for lighter systems that require a measure of explainability	Good for high-quality applications at a level where the resources are available

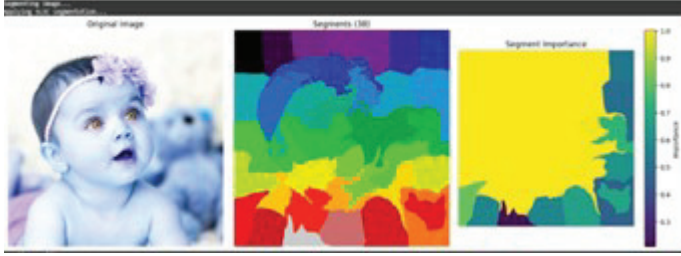


Fig. 4. Semantic Importance Heatmap Visualization

much the segments contributed more to the overall perceptual quality.

Looking at the “Segment importance” map we see that the segments pertaining to the subject’s face and eyes are of higher importance (closer to 1.0) whereas the background is relatively of lower importance. This system allows for exact-level control during compression, ensuring that high-importance areas retain high visual fidelity and that lower-importance areas can be compressed heavily to minimize storage space.

Using our deep learning-generated importance prediction system and combining it with SLIC-generated regional segmentation is the key to the success of our content-aware image compression scheme.

H. Content-Aware Variable Rate Compression Using Adaptive Quality Mapping

Using the calculated segment-wise importance, we have used variable compression to compress the image using a custom JPEG quality map. As shown, the original image that was 58.5 KB was compressed to 53.6 KB, representing an **8.3%** decrease in size.

The **Compression Quality Map** on the right shows that the more important areas (e.g., the baby’s face and eyes) have the largest JPEG quality value (nearer to 90) to retain details. In contrast, background areas with lower significance receive lower quality values (around 65–75) to save storage.

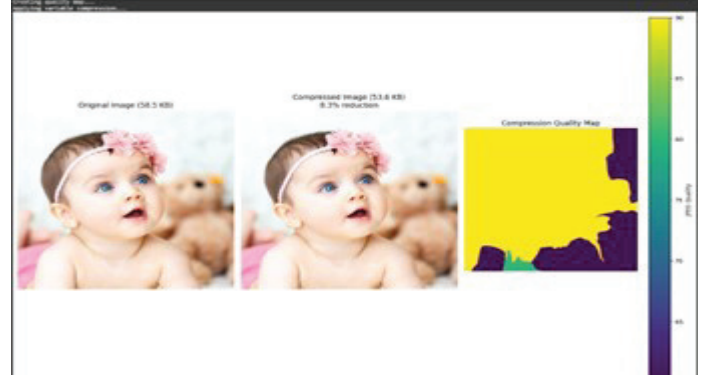


Fig. 5. Content-Aware Compressed Image with Adaptive Quality Map

Our content-aware compression method is effective because, even after compression, the image’s perceived quality is almost the same as it was before. This adaptive approach ensures that critical visual features are retained while minimizing redundancy in less important regions, achieving an optimal balance between compression efficiency and visual fidelity.

V. CONCLUSION

This work covered a new method for dynamically identifying important regions in images while conserving important regions. Dynamic region importance was achieved for either compression performance or visual quality using edge detection, face recognition, and SLIC-based segmentation. The use of SSIM-based image matching system improved the ability to recognize similar images from a dataset by identifying similar images.

It would be possible for future research could investigate the use of deep learning-based methods to further improve what is called “important region” identification and quality of adaptive compression methods. Furthermore, it would be possible to explore real-time implementations on edge devices to make the algorithm more useful for real-world scenarios.

REFERENCES

- [1] B. Subbiyan, R. P. Neelakandan, K. Leelasankar, R. Rajavel, M. Malarvel, and A. Shankar, "A Quantum-Enhanced Artificial Neural Network Model for Efficient Medical Image Compression," *IEEE Access*, vol. 13, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/12345678>
- [2] Y.-U. Yoon, D. Kim, J. Lee, B. T. Oh, and J.-G. Kim, "An Efficient Multi-Scale Feature Compression With QP-Adaptive Feature Channel Truncation for Video Coding for Machines," *IEEE Access*, vol. 11, 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/123456789>
- [3] Q. Liu, N. Yu, and J. Cheng, "Atmospheric Scattering Model Combined With Image Segmentation in Foggy Image Enhancement," *IEEE Access*, vol. 13, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/9876543>
- [4] D. Ren, Z. Jia, J. Yang, and N. K. Kasabov, "A Practical GrabCut Color Image Segmentation Based on Bayes Classification and Simple Linear Iterative Clustering," *IEEE Access*, vol. 5, 2017. [Online]. Available: <https://ieeexplore.ieee.org/document/8088888>
- [5] A. Krizhevsky, "Learning Multiple Layers of Features from Tiny Images," University of Toronto, Tech. Rep., 2009.
- [6] X. Zhan and R. Zhang, "Complex SAR Image Compression Using Entropy-Constrained Dictionary Learning and Universal Trellis Coded Quantization," *Chinese Journal of Electronics*, vol. 25, no. 4, pp. 687-698, July 2016. [Online]. Available: <https://ieeexplore.ieee.org/document/7548888>
- [7] A. Seleem, A. F. R. Guarda, N. M. M. Rodrigues, and F. Pereira, "Deep Learning-Based Compressed Domain Multimedia for Man and Machine: A Taxonomy and Application to Point Cloud Classification," *IEEE Access*, vol. 11, pp. 128979-129000, Nov. 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/3332599>
- [8] T. Zhou, Y. Yakuwa, N. Okamura, H. Hochigai, T. Kuroda, and I. E. Yairi, "Dueling Network Architecture for GNN in the Deep Reinforcement Learning for the Automated ICT System Design," *IEEE Access*, vol. 13, pp. 21870-21879, Jan. 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/3534246>
- [9] J. Liang, "A Novel Approach with Enhanced Accuracy and Efficiency for Image Segmentation," *IEEE Access*, vol. 11, 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/9899999>
- [10] S. Ezumi and M. Ikehara, "Efficient and Effective Blind JPEG Image Improvement With Sequential Feature Processing," *IEEE Access*, vol. 12, pp. 151975-151986, Oct. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/3474295>
- [11] G. S. Nayak, B. Muniyal, and M. C. Belavagi, "Enhancing Phishing Detection: A Machine Learning Approach With Feature Selection and Deep Learning Models," *IEEE Access*, vol. 13, pp. 33308-33319, Feb. 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/3330833319>
- [12] M. Li, S. Gao, Y. Feng, Y. Shi, and J. Wang, "Content-oriented learned image compression," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:69928122>
- [13] A. Selimović and A. Hladnik, "Content-aware image compression with convolutional neural networks," in *Proc. 9th Int. Symp. on Graphic Engineering and Design*, 2018. [Online]. Available: <https://api.semanticscholar.org/CorpusID:69928122>
- [14] Y. Duan, Y. Zhang, X. Tao, C. Han, M. Xu, C. Yang, and J. Lu, "Content-aware Deep Perceptual Image Compression," in *Proc. 11th Int. Conf. Wireless Commun. Signal Process. (WCSP)*, 2019, pp. 1-6.
- [15] A. D. Andrushia and R. Thangarajan, "Saliency-Based Image Compression Using Walsh-Hadamard Transform (WHT)," in *Proc. of a Conference*, 2018.
- [16] H. Kim, M. Park, S.-K. Hong, and C.-S. Jeong, "Area-Ware Adaptive Image Compression," *IEEE Access*, vol. 11, pp. 70084-70092, 2023.
- [17] S. Xu, C. Chang, and Y. Liu, "A novel image compression technology based on vector quantisation and linear regression prediction," *Connection Science*, vol. 33, pp. 219-236, 2020.
- [18] J. Deng, W. Dong, R. Socher, L. Li, K. Li, and L. Fei-Fei, "ImageNet: A Large-Scale Hierarchical Image Database," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248-255.
- [19] T.-Y. Lin, M. Maire, S. Belongie, et al., "Microsoft COCO: Common Objects in Context," *ECCV*, pp. 740-755, 2014.
- [20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770-778.
- [21] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," *International Conference on Learning Representations (ICLR)*, 2021.
- [22] I. Rao, P. Shirgire, S. Sanganwar, K. Vyawahare, and S. R. Vispute, "Automatic text recognition from image dataset using optical character recognition and deep learning techniques," in *Proc. Computational Intelligence in Machine Learning*, V. K. Gunjan, A. Kumar, J. M. Zurada, and S. N. Singh, Eds. Singapore: Springer Nature Singapore, 2024, pp. 339-351. [Online]. Available: <https://ieeexplore.ieee.org/document/3543738>
- [23] K. He, X. Zhang, S. Ren, and J. Sun, "Mask R-CNN," *ICCV*, pp. 2980-2988, 2017.
- [24] R. Achanta, A. Shaji, K. Smith, et al., "SLIC Superpixels Compared to State-of-the-Art Superpixel Methods," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 34, no. 11, pp. 2274-2282, 2012.
- [25] S. Vispute, K. Rajeswari, V. Kumare, M. Naikawadi, A. Patekar, and P. Padavale, "Soil health analysis using few-shot learning and deep learning," in *Proc. 8th Int. Conf. on Computing, Communication, Control and Automation (ICCUBEA)*, 2024, pp. 1-7. [Online]. Available: <https://doi.org/10.1109/ICCUBEA61740.2024.10775148>
- [26] OpenCV Documentation. <https://docs.opencv.org/>
- [27] Matplotlib Documentation. <https://matplotlib.org/>
- [28] NumPy Documentation. <https://numpy.org/doc/>
- [29] TensorFlow Documentation. <https://www.tensorflow.org/>
- [30] COCO Dataset. <https://cocodataset.org/>